

Monitoring Cognitive Performance with a Serious Game

A Longitudinal Case Study on Online Cognitive Assessment Using Serious Games

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ABSTRACT

This case study involves: the design and evaluation of serious games; the use of longitudinal research and remote testing in an international setting. Current methods for cognitive assessment tend to be inconvenient, costly and infrequently performed. This is unfortunate because cognitive assessment is an important tool. In the young it can detect atypical development, and in older people it can detect cognitive decline. For both young and old, cognitive assessments can identify problems and trigger interventions for reducing harms (e.g., adverse reactions to drugs) or providing treatment. Serious games for cognitive assessment can potentially be self-administered and played on an on-going basis so as to track cognitive status over time, something that is not practical with current methods. Inspired by this opportunity the BrainTagger team has developed a suite of cognitive assessment games. Studies are being carried out to assess the validity of these games for measuring the cognitive functions that they target, but those studies don't address the issue of whether people will be willing to play the game repeatedly, without supervision, over an extended period of time. Thus we carried out a longitudinal study with BrainTagger. We report on the logistical challenges of running this study with an international team located in Canada and Japan during the COVID19 pandemic. We also report on how the perceived "fun" of games changed over time. Our games were all versions of Whack-a-mole games, with each game requiring a different cognitive function to distinguish between targets (moles to hit) and distractors (moles to avoid). While the basic Whack-a-mole game is fun to play, having to play the same games again and again over a larger time period appeared to be more challenging than anticipated and motivation and acceptance seemed to gradually decrease over the course of the study. We conclude that addition of gamification features, such as leaderboards and in-game rewards, are needed to sustain enjoyment of our BrainTagger games and likely other games as well.

CCS CONCEPTS

• **Human-centered computing**; • **Human computer interaction (HCI)**; • **HCI design and evaluation methods**; • **User studies**;

KEYWORDS

Serious games, Cognitive assessment, Gamification, Evaluation, Longitudinal study

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1 INTRODUCTION

Many industrialized nations are confronted with a rapidly aging society and an accompanying increase in diseases associated with cognitive deficits (e.g. Dementia). In 2018 the World Health Organization reported that 50 million people worldwide have Dementia and the number is expected to grow up to 82 million by 2030 and 152 million by 2050 [18]. Unless new strategies for prevention and management are developed, this syndrome is expected to place growing demands on health and long-term care providers as the world's population ages. Cognitive assessment may be useful throughout the lifespan as a way to detect threats to cognitive status which might then be addressed by managing relevant lifestyle factors. During midlife there may be subtle changes in cognitive status due to conditions such as hypertension (high blood pressure) and poor glycemic control (e.g., in poorly managed diabetes). The assessment of cognitive functions in older people is of particular interest as those functions are related to the ability to maintain independence and to perform activities of daily living [5]. Cognitive assessment is also useful in children to detect if they are developing atypically, or are in need of educational enrichment. Thus assessments can be useful in detecting problems or opportunities relating to cognitive status at any age. The study reported here sampled young adults since young adults are typically experienced gamers and thus have a good understanding of what makes a game fun.

Performance on serious games can be used to predict, diagnose and possibly guide treatment of changes in cognitive status [13]. Serious games are a promising technological approach to cognitive assessment because they make assessment entertaining and easy to perform, and they enable a more cost-effective and frequent cognitive assessment. In order to serve as an early detection tool serious games should be capable of long-term monitoring of cognitive performance. In the international case study reported here, our Serious Game was tested as a monitoring tool of cognitive performance over a period of five weeks with students. Due to the emergency situation brought on by the COVID19 pandemic, all interactions with the participants took place online, which provided the opportunity to test the game for remote access in a solely online environment.

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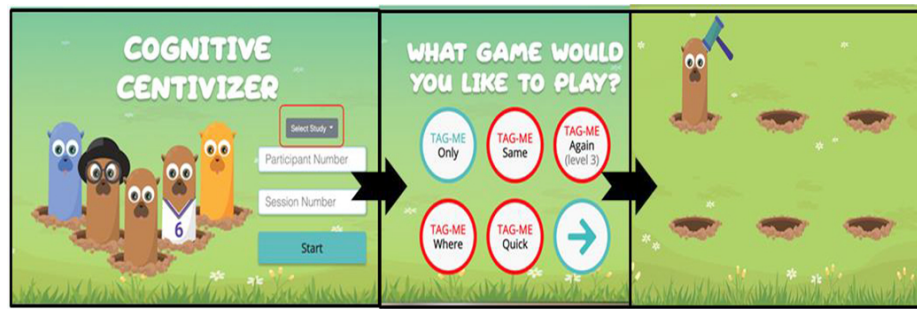


Figure 1: Screenshots of the Whack-a-mole game used in the study

The game developer and project manager were located in Toronto, the principal investigator was in Tokyo and the participants came from both Tokyo and Europe. In this case study, we report on the experiences of the participants and the problems that they encountered. Based on our findings we make recommendations concerning the design of Serious Games as healthcare monitoring tools.

1.1 Serious Games

Serious games are games that have an underlying purpose other than entertainment. Ideally, serious games should be fun and they should motivate target users to carry out tasks that they would otherwise find difficult or boring in settings such as education or healthcare [17]. There is empirical evidence that serious games have a positive impact on affect and motivation, increase knowledge acquisition, and improve perceptual and cognitive skills [3, 8]. Serious games engage the player in a fun and entertaining way [1] and keep users engaged in health behavior change [7]. Fleming et al. [4] carried out a systematic review for serious games and identified six main types of games; sport or movement-based games, virtual reality games, serious games for cognitive behavior therapy, entertainment games, biofeedback-based games and cognitive training games. In these games players have to defeat characters in the game [10, 11] or compete against other players [6, 11]. These games typically have an achievement component where players progressively improve their skill level [12] and level-up [6, 11]. However, the purpose of our serious game [14] differs from these games by focusing on the valid and reliable assessment of cognitive performance and not its improvement. While the games mentioned above focus on knowledge acquisition, behavior change, or skill training, our game is designed to assess a person's cognitive functioning in a reliable and valid way that allows for repeated assessments over an extended period.

Since 2014 we have been developing serious games for cognitive assessment based on the popular Whack-a-mole game. In the standard version of the game, moles appear randomly from different holes and the player has to hit each mole as quickly as possible. Our games building on this basic idea and vary in complexity, but the simpler games can be played by people with cognitive impairments and dementia. Sample screenshots from the current suite of games is shown in Figure 1. In previous studies we showed that the game can be used to screen for Delirium in Emergency patients [9], and that game performance correlates with the Mini-Mental State

Examination, a widely accepted measure for cognitive impairment [15].

1.2 Research Questions

Despite claims concerning the benefits of using serious games, evidence from long-term studies is scarce [2]. A major concern is how motivated users will be to play the same games regularly over a period of weeks or months. Our game differs from other serious games by requiring the player to carry out the same action (hitting the mole) repeatedly with having no or little opportunity for improvement, a necessity to collect sufficient data to measure cognitive performance reliably. Different types of games in our gaming environment with different rules offer some variety, but is this enough to motivate players? In addition, given the pandemic happening during this study, we wanted to know if the games could be played online for an extended period of time, and without supervision. The question addressed in our research were therefore:

RQ1: Will players be willing to play our whack-a-mole games repeatedly over the course of five weeks and continue to enjoy playing the games?

Because of the COVID situation the entire study had to be carried out online and face-to-face meetings with participants of the study were not possible. Therefore a second research question evolved:

RQ2: Can our serious game be used for repeated measurement of cognitive performance carried out online by the user and without facilitation?

Since this research was an international collaboration our third research question was:

RQ3: What challenges arise from conducting a study with an international research team working from different continents?

2 METHOD

2.1 Participants

We recruited 19 students (ranging from 21 to 25 years of age, 10 female) from an online class at Keio University (Japan) to participate in the study.

2.2 Material

Table 1: Seven different games used in the Case Study

Game	Cognitive function	Definition	Game play
TAG-Me Quick	Cognitive speed	Speed of task performance	Moles pop out randomly from one of six holes. Players have to hit moles as quickly as possible.
TAG-Me Only	Response inhibition	Suppress inappropriate actions	Avoid hitting distractor moles (moles wearing a hat) while hitting regular moles as quickly as possible
TAG-ME Again	Working memory	Holding and memorizing information temporarily for completing a task	Remember a number on the T-shirts the moles are wearing. Hit a mole wearing a T-shirt with the same number as a mole appearing three positions back.
TAG-ME Switch	Dimensional shifting	Cognitive flexibility to switch between arbitrary rules	Six rules of what kind of mole to hit are randomly switched during game play. E.g., moles with the same colors have to be hit, or moles appearing on the left hand side, etc.
TAG-ME Where	Spatial attention	Awareness of the physical environment	A lightning bolt appears above specific holes indicating a hole with higher probability where the next mole will appear .
TAG-ME Same	Visual short-term memory	Memorizing visual information for a short period	A target mole varying in color is shown before a mole appears from a hole. The mole matching the target has to be hit as quickly as possible.
TAG-ME Bigger	Inhibition of processing conflict	Inhibiting responses to irrelevant size feature	Moles varying in size wearing T-shirts with numbers on it. The mole with the larger number on the T-shirt has to be hit regardless of the mole's physical size.

2.2.1 Serious Game: Whack-a-Mole. BrainTagger is a suite of Whack-a-mole TAG-ME (Target Acquisitions Games for Measurement and Evaluation) games for cognitive training and assessment. The BrainTagger games used in this study are summarized in Table 1. Performance measures included speed and various forms of accuracy.

2.2.2 Game Interface. An important aspect of the study design was to ensure that games are played in randomized order to control order effects. In consultation with the PI, the development team created a special game plug-in making sure games could only be played in a predefined randomized order. The order of the games to be played was indicated by color circles (grey - already played, blue - next game to be played, red - must be played later) around the game buttons, as shown in Figure 2

2.3 Procedure (Project Flow)

The project went through three phases: 1) preparation; 2) data collection, and 3) evaluation.

1. **Preparation** (one week): Detailed written instructions on the scope of the project, the project procedure and game instructions were sent to the participants. Participants received a link to a demo version to familiarize themselves with the game interface and rules. In an online session questions could be discussed.

2. **Data collection** (four weeks): Over the course of four weeks participants were asked to play all the seven games (see Table 1) once per week. Games were randomized across players in all four sessions. Responses were made by hitting six keys which were each mapped to a different hole in the game. Participants received regular feedback about task completion. Participants could ask questions in weekly online sessions.
3. **Evaluation phase:** Participants prepared a short report detailing their experience playing the game, making suggestions for game improvements.

3 RESULTS

3.1 Participation Rate

Only eight of the 19 participants completed the study in full. Seven of the remaining participants completed three sets of games (i.e., 15/19 of the participants completed at least 75% of the requirement), one participant completed two sets, two participants completed one set, and one participant only did one incomplete set of games. Detailed feedback experience with the games was received from 17 participants.

3.2 Server End Problems

Various technical problems occurred while running the study. Most of the technical problems related to sending data to the server. Some participants reported that they played the games, but the back-end



Figure 2: Game interface, colored circles indicate the order in which the games had to be played

server did not receive any data from them. After troubleshooting we found out that browsers other than Chrome, such as Firefox and Safari, had individual privacy settings preventing the games from sending data.

In addition, the back-end servers failed twice during the four weeks of data collection. Analysis of server logs showed that the CPU capacity of our virtual server was exceeded when participants tended to bunch up and play games at the same times, more than was anticipated (based on lessons learned in this study the servers have since been upgraded).

3.3 User End Problems and Suggestions for Improvement

3.3.1 Technical Problems. About half of the participants ($n = 10$) reported technical problems such as problems with sending data ($N = 3$), loading the moles for the game ($n = 4$), and connection problems ($n = 3$). Two participants specifically mentioned that they were very frustrated by repeatedly encountering technical problems.

3.3.2 Problems with Game Play Instructions and Game Interaction. The majority of participants noted that the games were too long ($n = 12$). Two participants felt that the instructions were too complicated and difficult to understand, but others complained that having to do a trial run for each game every time was too boring and time consuming. Three participants reported that having to use keyboard keys limited the motions involved in playing the game and felt it was difficult to play the game using only keyboard keys. Five participants found the game background music annoying or boring. Participants suggested that the player should have control over the background music played ($n = 7$), or that different background music should be played for each game ($n = 3$).

3.3.3 Enjoyment and Motivation. Half the participants found the games to be repetitive ($n = 9$), reducing their motivation to play the game. The games were seen as being too simple and not exciting enough ($n = 3$). While participants in the initial online sessions mentioned that playing the games was fun and entertaining, having to play the game repeatedly over several weeks created feelings of boredom and frustration for most of the games.

Most of the participants also mentioned scoring in their feedback report ($n = 12$). Participants stated they kept tracking their scores during game play and expressed frustration when not being able to improve their scores ($n = 6$). One player tried different approaches for playing the game to find out how to improve the score.

3.4 Reflections from the Principle Investigator (PI)

In addition to the development team, which was located in Toronto, a Tokyo-based PI maintained contact with the participants, sent instructions, answered questions and held weekly online meetings, which were voluntary. A big problem from the start of the study was that participants did not read and follow written instructions carefully. The instructions were carefully drafted, having step by step explanations written in simple language with accompanying screenshots. Many questions raised in the online meetings referred to when, how often, and how many of the games had to be played. Several participants realized through our discussions in the online meeting that they should only use the keyboard keys and no other input method. Since the study was carried out during the COVID19 pandemic when all classes were held online, participants struggled with managing their time schedule and incorporating game play into their weekly routine. Constant reminders by the PI were necessary to ensure that data over all four sessions could be collected. Without the weekly online meetings the drop-out rate might have been even higher. The online meetings added a personal touch that

helped to motivate participants in spite of the technical difficulties, and boredom, that some participants experienced.

3.5 Working as an International Research Team

Our international team had weekly meetings at the beginning of the project. These meetings were very helpful in creating a common understanding of the goals of the project and how it should be carried out. Throughout the data collection phase, the PI received regular updates from the developer team on data collection and participation rates; so that the PI was able to address any issues that arose in the weekly online sessions with the participants, explaining the background and causes of the technical difficulties to promote participant understanding and buy-in. The smooth communication between the project members was key to the successful completion of the project. Because of the time difference of 13 hours between Toronto and Tokyo (the study was conducted during Eastern Daylight Time), we agreed on a specific day and time for the meeting at the beginning of the project and kept to that schedule.

Participants based in Tokyo could easily participate in the weekly online sessions with the PI, but this was not possible for participants from Germany and the UK, making it difficult to address their problems.

4 DISCUSSION

We conducted a longitudinal study over the course of five weeks to test a serious game for assessing cognitive performance. The detailed feedback received from participants after the study helped us to understand the difficulties and challenges they experienced. The research questions are addressed below and suggestions for how to address the challenges identified are provided in the implications section.

RQ1: How motivated are players and how acceptable find players to play our Whack-a-mole game repeatedly over the course of five weeks?

When we developed the game we believed the games to be fun to play, the characters in the game were designed to be cute and initial studies indicated that one of the games was well received with over 95% compliance when it was offered to older emergency patients [16]. However, having to play the same games again and again over a larger time period appeared to be more challenging than anticipated and participants reported that the games were monotonous. Motivation and acceptance seemed to gradually decrease over the course of the study. Participants focused a great deal on the scores they got when playing the game. This suggests that improved scoring and feedback procedures may be key to motivating continued game play. We suggest that overall performance in different games should be presented in graphs combining performance across several sessions so that players can identify trends over time, and thus better understand what their strengths and weaknesses are. Incentives to play can be used as an additional motivator. Further motivation may be provided with a timer giving feedback on how much playtime is left, and with player control of background music. We also need to revisit the timing of game events, and the use of gamification features to motivate users to keep playing the games.

RQ2: Can our serious game be used for repeated measurement of cognitive performance carried out online under the control of the user?

We were able to collect complete data sets from 42% of the participants and datasets with only one week of data missing from a further 37% of the participants. Thus there was fairly good compliance from around 80% of the participants, but with additional adjustments and changes we can probably increase the participation rate. The game needs to be improved in how instructions are delivered to make sure that participants use the intended input method and understand when and in what frequency the games have to be played. A video demonstration is probably more appropriate especially for younger players considering their media habits. Additionally, instruction videos might give participants a better idea in which order they have to follow instructions and also provides a demonstration of the game play. Repeatedly occurring technical problems were a hindrance in the project that also negatively affected participants' motivation. Having a downloadable form of the game that can be used offline, collects data and sends it to the server once the user goes online would avoid many problems regarding users' internet connection and loading difficulties. In addition, the game should include features that help players manage and keep track of their game schedule by sending automatic reminders and monitoring their weekly performance.

RQ3: What challenges arise from conducting a study with an international research team working from different continents?

Probably one of the biggest challenges was to keep everyone involved in the project up to date on progress and to react quickly and without major delays. Due to the large time difference between Tokyo and Toronto (13 hours), we planned online meetings only once a week and communicated mainly via e-mail. Having a PI in the same place as the participants was a great advantage. The PI was able to react quickly to questions or requests from the participants, collected complaints and sent them as a summary to the developer team. With a regular flow of information, an international research team does not work much differently than a local team. The biggest challenge was probably keeping in touch with the participants from Europe. Due to the large time difference with Japan, these participants were unable to attend the weekly online meetings with the PI. Although we did not measure the level of frustration or satisfaction with the game, our impression was that the participants who did not attend the online meetings had the most complaints and a higher dropout rate compared to the participants who attended the online meetings.

This study tested the use of our serious games over a period of five weeks, but ideally the studies should be continued over a longer period of at least one year. The small number of about 19 participants may limit the validity of our results. The participants in our study were young adults, as they are usually experienced players and easier to study than children or older people, but the results of our study need to be validated with other age groups.

5 IMPLICATIONS

5.1 How to Avoid Drop-Out

Drop-out is a major problem especially for online studies. We believe three approaches can significantly reduce dropout such as (1) performance feedback, (2) incentives, and (3) personal monitoring/mentoring.

1. **Performance feedback:** Game players want to monitor their own performance. Sophisticated ways of performance feedback that go beyond simply displaying game scores can motivate players to continue playing a game in the long term. Important aspects of such feedback include: showing how performance changes over time across different game parameters, providing feedback on one's own performance improvements and in comparison to others, highlighting individual strengths and weaknesses.
2. **Incentives:** Incentives (e.g., financial, or in-game rewards) can motivate players to play a serious game regularly if they are conditional on regular play. Incentives can be of monetary value or new game options. In-game rewards can include unlocking new game characters, dressing game characters, unlocking different musical themes, etc. Participants should be able to choose the type of incentive. Additional incentive should be provided for completing a block of sessions.
3. **Personal monitoring:** Probably one of the best ways to motivate people is support and encouragement from others. Communicating regularly with the player, being open for conversation, responding to players' feedback, and offering encouragement (but also understanding the challenges a player faces), is an important part of building a connection between player and the agency that collects the data for health monitoring. When the Serious Game is played by older people, family members should be involved, to provide reminders and to encourage them to play the game regularly.

5.2 What to do about Technical Problems

Based on the experience gained in our project with technical problems, we make the following suggestions: (1) quick feedback when problems occur, (2) checking individual user settings, (3) providing alternative solutions.

1. **Quick feedback:** Quick feedback is crucial to gain user confidence and to avoid declining motivation. Even if the problems persist and cannot be solved immediately, the team must at least show understanding for the user's situation, communicate that the team is working on the problem, and inform the user of the approximate time it will take to solve the problem.
2. **Checking the individual user settings:** Users may have blockers or add-ons installed in their browser that may interfere with the game and data collection. It is difficult to foresee this kind of problem which needs to be solved by working directly with the user either by using screen sharing to observe a session, or by requesting screenshots to investigate the problem.
3. **Providing alternative solutions:** If the problems persist and none of the previous steps have worked, alternative

solutions should be provided to users. For example, instead of playing an online version, offer a downloadable version of the game or provide the user with a specific device to play the game.

5.3 How to Avoid Player Fatigue

In monitoring applications, games are played over extended periods. Since the games use repeated actions, gamification techniques are needed such as leaderboards and badges to ensure that players remain motivated to play the games. Our suggestions for reducing player fatigue are: (1) variation in interaction methods, (2) diversifying game play, (3) incorporating learning opportunities.

1. **Variation in interaction method:** Providing more alternative interaction methods beside keyboard keys to motivate active motions and movements. For instance, we have developed a button box (as demonstrated in the video) that can be used for game play.
2. **Diversifying game play:** The game environment can be enriched by adding games that have mainly entertainment value and provide an entirely fun and positive play experience. Embedding the "serious game" into such an environment or allowing players to switch between "serious game" and "entertainment game" could provide welcome refreshment between measurements and reduce fatigue.
3. **Incorporating learning opportunities:** Players are motivated by their success, seeing improvements in their scores or reaching a new play level. The game environment should include games that provide learning opportunities and give the player a feeling of success. However, additions to the game must not compromise its validity in assessing the targeted cognitive abilities.

6 CONCLUSION

This case study reported on a longitudinal study of a suite of serious games for cognitive assessment. The games were remotely tested and self-administered by participants in an international setting. Using a serious game for cognitive assessment is a convenient and cost-effective way to detect subtle changes in cognitive status at any age. The main problems identified in the study were how to increase motivation and enjoyment when the same game has to be played repeatedly, how to address technical problems encountered by the end-user, and how to ensure that instructions are followed. The study showed that feedback received in the longitudinal study covered aspects that had not previously been considered in the design of the game and underlines the importance of conducting this type of evaluation. Serious games are developed for a specific purpose, but factors that influence player fatigue and motivation must be considered to increase acceptance and willingness to play these games regularly. This study lasted five weeks, but ideally studies should be continued over a longer period of at least one year and should use a larger sample of participants.

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